

Cover Letter

Subject: Application — Cheminformatics Engineer / Scientific Software Developer (BioSpecInfo project)

Dear [Company] Hiring Team,

my name is **Samuele Pio Provenzano** and I am a Chemistry graduate from the University of Bari Aldo Moro. I combine a solid background in organic chemistry, spectroscopy and physical chemistry with the ability to **design and build scientific software** using modern web technologies.

I independently designed and developed **BioSpecInfo** (<https://samupropio1-ship-it.github.io/BioSpecInfo-v11/>), a Progressive Web App for cheminformatics, 2D/3D molecular modelling and spectral prediction, with a few distinctive features:

- **Local-first architecture:** a ≈ 4.5 MB application, no backend, 100% offline, powered by the **RDKit engine compiled to WebAssembly**.
- **Validated spectral predictors:** benchmarked against official databases with a mean absolute error of **0.31 ppm** ($^1\text{H-NMR}$ vs SDBS) and **11 cm^{-1}** (IR C=O band vs NIST WebBook).
- **Broad coverage:** Spectra Centre (IR, NMR, MS with isotope pattern, UV-Vis, Raman), 46 retrosynthetic strategies with animated 2D/3D mechanisms, 25 organic chemistry modules, drug-likeness/ADMET (Lipinski, Veber, QED, Brenk/PAINS alerts).

I believe my **dual expertise — advanced chemistry and software engineering** — can bring concrete value to your R&D, molecular modelling or digital health projects, closing the gap between those who understand chemistry and those who build the tools to analyse it.

I attach my CV and the **full technical dossier** of the software (architecture, validation, security and licensing). I would be glad to walk you through the project in an interview or with a short live demo.

Thank you for your consideration.

Kind regards, **Samuele Pio Provenzano** ✉ samupropio1@gmail.com · 📱 [phone] · 🔗 [LinkedIn]